Phase 2: Literature Review

NutriVision

Introduction:

This literature review examines existing dietary health applications, the role of artificial intelligence in nutrition, technical frameworks, and UI/UX best practices. It identifies gaps in current solutions and establishes the foundation for NutriVision.

1. Dietary Health Applications:

Mobile health (mHealth) apps such as MyFitnessPal and Yazio have improved diet quality and user engagement. Systematic reviews confirm that smartphone apps consistently enhance healthy eating behaviors in adults . Gap: Most apps rely on static databases and manual input, with limited personalization or AI-driven recipe generation.

2. AI in Nutrition & Healthcare:

Artificial intelligence has emerged as a transformative force in nutrition and dietetics, enabling personalized diet plans, monitoring food safety, and supplement recommendations . Opportunity: NutriVision leverages prompt engineering to generate tailored recipes and suggest healthier alternatives, addressing gaps in personalization.

3. Technical Frameworks:

- Flutter & Dart: Provide cross-platform development with responsive UI.
- Firebase Cloud Firestore: Enables real-time synchronization across devices, widely adopted in mobile health apps .
- Dio Package: Offers secure API communication with interceptors, and can integrate with Firebase performance monitoring .

4. Prompt Engineering in AI:

Prompt engineering is recognized as a critical skill for guiding large language models to produce accurate, relevant outputs. It is increasingly applied in healthcare and content generation . NutriVision applies prompt engineering to dietary contexts, enhancing recipe generation and snack alternatives.

5. UI/UX Best Practices:

Healthcare app design emphasizes empathy, accessibility, and intuitive interfaces. Research highlights the importance of design thinking, wireframes, and user personas to improve adoption and trust in health applications . NutriVision follows these principles to ensure a user-friendly experience.

6. Research Gaps:

- Lack of AI-driven personalization in mainstream dietary apps.
- Limited use of real-time synchronization for multi-device tracking.
- Few solutions combine AI prompt engineering, secure networking, and responsive UI/UX in one integrated system.

Conclusion:

The literature confirms the need for NutriVision as a next-generation dietary health assistant. By integrating AI-driven nutritional analysis, prompt engineering, real-time sync, and modern UI/UX, NutriVision addresses gaps in existing solutions and aligns with best practices in health technology.

Appendix: References documented in APA 7th edition style:

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